Singular companions shared by the skipped and published curves

Date: 2026-10-23 UTC. **Scope:** Exact algebraic audit of the eleven reconstructed omitted seed classes and eight published controls over the same four prime fields. **Result:** This specific formula-compatible input surface is identical within each field and cannot separate the published and omitted labels.

Question and exact result

Under the working premise that the original program evaluated all eleven omitted curves and rejected them for a weakness, could it have avoided an implementation weakness involving **singular points**? For the usual short-Weierstrass addition formulas, a scalar routine that accepts unvalidated points can operate on other equations with the same `a=-3`. Two of those equations are singular: $y^2=x^3-3x+2$ and $y^2=x^3-3x-2$. Their smooth points form finite groups of order $p-1$ or $p+1$, determined by the field prime and a quadratic-character check.

Derived and independently checked: All nineteen named curves have the same `p` and `a=-3` within each field. Therefore both singular cubics and their smooth-point orders are the same for every omitted and published candidate in that field. This excludes a proposal in which the omitted coefficient `b` uniquely exposes these singular inputs while the published `b` avoids them. The result is conditional on an implementation that skips point validation and whose reached formulas omit `b`; no such historical or deployed implementation was examined. These singular cubics are neither elliptic curves nor isogenous representatives of the source curves.

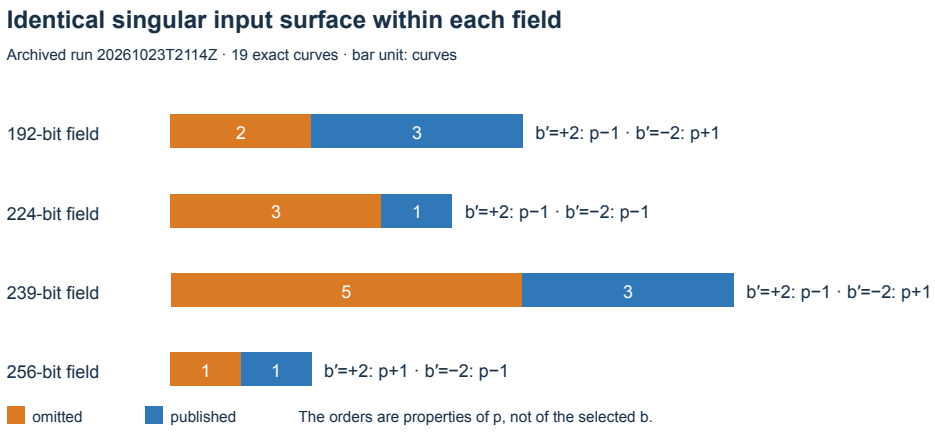


Figure 1: Exact field comparison

Figure 1. Exact counts for the nineteen UIDs in the sealed primary record. Each bar unit is one seed-level curve; no statistical uncertainty attaches to these finite counts. Orange is omitted and blue is published. The p±1 labels are exact smooth-locus orders for the two singular inputs. Editable source: draw.py.

Field	Omitted	Published	$b'=+2$ smooth order	$b'=-2$ smooth order
192-bit prime	2	3	$p-1$	$p+1$
224-bit prime	3	1	$p-1$	$p-1$
239-bit prime	5	3	$p-1$	$p+1$
256-bit prime	1	1	$p+1$	$p-1$

Why there are exactly two singular companions

For $y^2=x^3+ax+b'$ with $a=-3$ over a prime field of characteristic greater than three, the nonsingularity condition is $4a^3+27b'^2 \neq 0$. Its failure gives $b'^2=4$, hence exactly $b'=\pm 2$. The factorizations are exact over the integers:

$$x^3-3x+2 = (x-1)^2(x+2)$$
$$x^3-3x-2 = (x+1)^2(x-2).$$

For $b'=+2$, put $u=x-1$; the equation becomes $y^2=u^2(u+3)$. For $b'=-2$, put $u=x+1$; it becomes $y^2=u^2(u-3)$. Outside the singular node $(u,y)=(0,0)$, slope $s=y/u$ gives $u=s^2-c$, with $c=3$ or -3. Every s except a root of $s^2=c$ gives one smooth affine point; add the smooth point at infinity. If c is a square there are two excluded slopes and the total is $p-1$; otherwise there are none and the total is $p+1$. This counts the smooth locus only; the node is never counted as a group element.

The frozen protocol pinned the two input JSON digests, all 19 EC1 UIDs, the discriminant and factorization checks, Euler-character checks, and two small-field enumeration controls before running. The primary record was generated from clean committed source `88750ee2d2` and has 19 rows, four field primes, and four passing small-field controls. Its `complete.ok` binds the result hash. The independent Sage record verified primality of each field with proof enabled, independently checked the polynomial identities and quadratic characters for every UID, and enumerated the smooth points at $p=101,103$ for both signs. All 19 UID checks and four enumerations passed, with a separate verification completion digest. The primary run was archived at commit `d07ee660cf`; software and host details are in both records. The source curve's ordinary group order and twist are unchanged by this analysis.

What this does and does not say

The prior nonsingular invalid-curve audit found that the b-independent arithmetic surface is shared within each field. This extension closes its deliberate singular-control gap by identifying and counting both singular cubics exactly. Neither audit tested a point decoder, scalar multiplication implementation, secret-key protocol, or output oracle. Even if an implementation processes a singular point, a useful key-recovery claim still needs a reachable smooth-locus point of suitable order, key reuse, observable outputs, and complete attack cost. Group size alone does not establish those conditions. For the historical selection question, a hidden criterion must therefore depend on something other than this p,a-only input surface, such as an actual b- or group-dependent mathematical property or implementation behavior supported by an original trace.

The exact identities and row counts are `internal` evidence in the linked immutable records; the smooth-point count is a `derived` algebraic result independently checked by Sage. The standard addition formulas' omission of `b` is visible in the Explicit-Formulas Database (`retrieved`, primary). The historical-evaluation premise remains stipulated, not observed; the selection-rule problem remains open. Existing canonical ECDLP graphs were checked; this result adds no verified attack edge, so none changes.

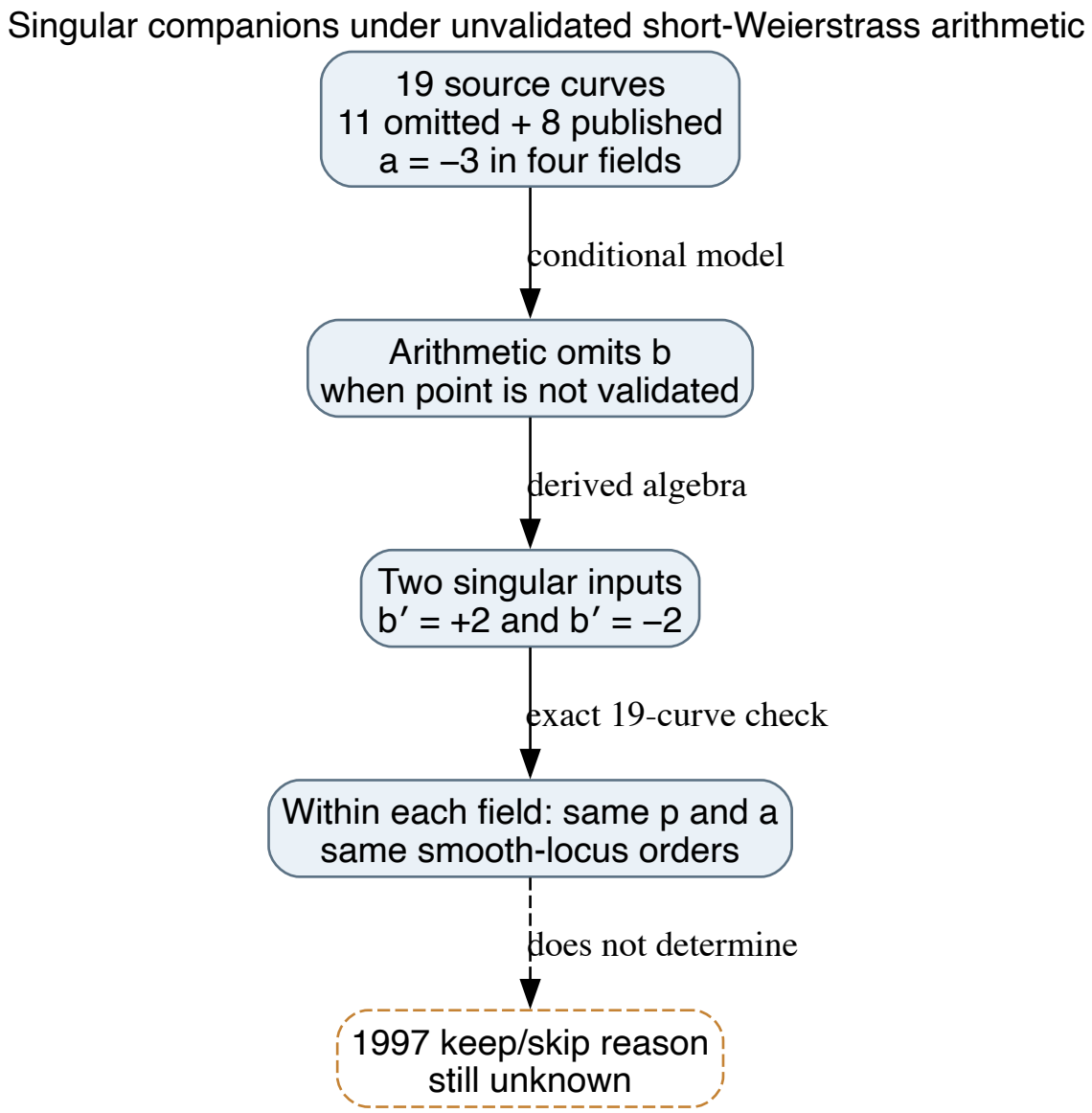


Figure 2: Inference boundary

Figure 2. Solid arrows are conditional algebra and exact checks; the dashed endpoint is the unobserved historical decision. Editable source: inference.dot.